

AN IMPROVED EXPLICIT LOWER BOUND FOR THE UNIT DISTANCE PROBLEM

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ABSTRACT. For a finite set $U \subset \mathbb{R}^2$, let $u(U)$ denote the number of ordered pairs of points in U at distance exactly 1. In this paper we improve upon the best existing explicit lower bound for the Erdős unit distance problem, due to Sawin [14], and prove that there are arbitrarily large finite sets $U \subset \mathbb{R}^2$ satisfying

$$u(U) \geq C^{-1}(\#U)^{1.0358324},$$

where $C > 0$ is an absolute constant. The construction follows the number field strategy introduced in the recent disproof of Erdős's unit distance conjecture, where unit distances arise from elements of prescribed relative norm in an ideal of a CM field, and the fields are supplied by a Golod-Shafarevich tower over a real quadratic field $\mathbb{Q}(\sqrt{D})$. The main new ingredient is a tower that is allowed to ramify at the unique dyadic prime of $\mathbb{Q}(\sqrt{D})$, which makes primes that split in $\mathbb{Q}(\sqrt{D})$ affordable in the Golod-Shafarevich criterion.

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1. INTRODUCTION

For a finite set $U \subset \mathbb{R}^2$, let

$$u(U) = \# \{ (u, v) \in U^2 : |u - v| = 1 \}$$

denote the number of ordered pairs of points of U at distance exactly 1, and let $u(n)$ be the maximum of $u(U)$ over all sets with $\#U = n$. In 1946, Erdős [3] proved that a $\sqrt{n} \times \sqrt{n}$ section of a suitably scaled integer lattice satisfies

$$u(n) \geq n^{1+c/\log \log n}$$

for an absolute constant $c > 0$, and conjectured that this lattice construction is essentially optimal [3, 4]. The best existing upper bound, due to Spencer, Szemerédi and Trotter [16], is

$$u(n) = O(n^{4/3}).$$

Determining the true order of $u(n)$ is the Erdős unit distance problem.

Erdős's conjecture was recently disproved. A team at OpenAI [13] constructed sets with $u(U) \geq (\#U)^{1+\delta}$ for a positive but inexplicit constant δ , replacing the integer lattice by ideal lattices in number fields of large degree and small root discriminant, with the fields supplied by class field towers. In a set of remarks on the proof, Alon, Bloom, Gowers, Litt, Sawin, Shankar, Tsimmerman, Wang and Wood [1] produced a simplified version of the argument, and obtained $\delta \approx 6.24 \cdot 10^{-38}$. Sawin [14] then reworked and optimized the construction, and proved that for arbitrarily large n ,

$$u(n) \geq n^{1.014114}/C.$$

In this paper we improve upon Sawin's exponent. Our main result is:

Theorem 1. *There exist an absolute constant $C > 0$ and arbitrarily large finite sets $U \subset \mathbb{R}^2$ such that*

$$\#\{(u, v) \in U^2 : |u - v| = 1\} \geq C^{-1}(\#U)^{1.0358324}.$$

In the notation above, Theorem 1 says that $u(n) \geq C^{-1}n^{1.0358324}$ for arbitrarily large n .

We follow the number field strategy of [14], which we now recall. Let K be a CM field of degree $2d$ with maximal totally real subfield F , let c denote complex conjugation, let I be a fractional ideal of K , and let $\alpha \in F$ be totally positive. Every solution of

$$\beta c(\beta) = \alpha, \quad \beta \in I,$$

satisfies $|\sigma(\beta)|^2 = \sigma(\alpha)$ at each complex embedding σ of K , and so after dividing the coordinate at the chosen embedding σ_v by $\sqrt{\sigma_v(\alpha)}$ at each real place v of F , each solution becomes a unit vector in $\mathbb{C}^d$. When I is realized as a lattice $\Lambda \subset \mathbb{C}^d$ under this normalization, the solutions become lattice vectors of length exactly 1 in a normalized norm, and projecting to a single complex coordinate is injective on K . Intersecting Λ with a ball of bounded radius and projecting then produces a finite planar set in which each norm solution generates many unit distance pairs.

The quality of the construction is governed by a ratio of two logarithmic quantities, a numerator measuring how many solutions β can be guaranteed, and a denominator measuring how many lattice points the ball may contain. Solutions are produced by a pigeonhole argument in a relative class group, from a finite set S of rational primes, the *selected primes*, whose primes in F split in K , weighted by multiplicities $k(p)$. The pigeonhole loses a factor of the relative class number $h^-(K) = h(K)/h(F)$, which is bounded by Louboutin's method. Finally, CM fields K of arbitrarily large degree with all of the required local properties are supplied by a 2-tower over a real quadratic field $Q_0 = \mathbb{Q}(\sqrt{D})$, and the infinitude of the tower is certified by a Golod-Shafarevich criterion.

All of this is quantitative: Theorem 10, our version of the master criterion of [14, Proposition 10], achieves every exponent below $1 + N(R)/M(R)$, with the numerator $N(R)$ and denominator $M(R)$ given by (5.1) and (5.2). In particular, for our certificate we have that $N(R)/M(R) > 0.0358324$ (section 7), with $N(R) \approx 37.887$ and $M(R) \approx 1057.33$ at the optimized radius R .

We stress seven differences between our construction and that of [14]. First, the number of points of Λ in a ball of normalized radius R is bounded by volume over covolume, uniformly in the center, using a lower bound for the minimum of the dual lattice and Banaszczyk's transference theorem [2] (Proposition 2). This replaces the packing bound $\#U \leq (2RA + 1)^{2d}$ of [14, Lemma 2] and shortens the denominator by roughly $\frac{1}{2} \log \lambda$ per unit of $2d$, where λ is the relative root discriminant, about 3.6 percent of the final denominator. Second, the overlap factor $(1 - \frac{1}{R})^{2d}$ of [14, Lemma 2] is sharpened to the exact two ball overlap $\Theta_{2d}(R)$ of Proposition 3, which loses only $\frac{1}{2} \log(1 - 1/(4R^2))$ per unit of $2d$ in the numerator rather than $\log(1 - \frac{1}{R})$, and together with the covolume count this permits the small radius $R \approx 2.69$ used by our certificate.

Third, the pigeonhole is performed in the group $G_{K/F}^+$ of ideals with totally positive principal relative norm, modulo a natural equivalence (section 3). For K/F quadratic and unramified at all finite primes, we have that $\#G_{K/F}^+ \leq 2h^-(K)$, whereas the group used in [14, Lemma 6] has order up to $2^{d+1}h^-(K)$, and so this is worth $\frac{1}{2} \log 2 \approx 0.35$ in the numerator. Fourth, in Louboutin's bound for $L(1, \chi_{K/F})$, the Dedekind zeta

function ζ_F appearing on the right is majorized, after normalization by degree, by ζ_E for any fixed subfield $E \subseteq F$ (section 4). Every field in our tower contains the fixed multiquadratic field $E_{39} = \mathbb{Q}(\sqrt{q} : q \in T, \sqrt{2})$, where T is the set of the first 38 odd primes fixed in section 7, whose splitting behavior is completely explicit, and so this yields $h^-(K)^{1/d} \lesssim \sqrt{\lambda} \cdot B_{E_{39}}(\lambda)/(2\pi)$ with $\log B_{E_{39}}(\lambda) < 0.09$ (section 7.1). The corresponding term in [14, Lemma 9] is $\frac{1}{2} \log \log \lambda \approx 2.17$ for our λ , and the refinement recovers all but 0.045 of it, which is decisive on the scale of the numerator $N(R) \approx 37.9$.

Fifth, the tower is permitted to ramify at the unique dyadic prime $\mathfrak{q}$ of Q_0 , in a controlled two dimensional direction, and a theorem of Schmidt [15] supplies an exact relation count for the resulting $\mathfrak{q}$ -ramified Galois group (section 6). Sixth, the relation counting is done in the weighted Golod-Shafarevich criterion of Vinberg and Koch [18, 9] rather than the unweighted criterion of [14, Lemma 11]: a dyadic reciprocity lemma shows that the second square Frobenius relation at each split selected prime has Zassenhaus degree at least 3 (the degree filtration is recalled in section 6), where relations are far cheaper, and so split primes become affordable in large numbers. This weighted count, with the second split prime relation placed at degree 3, was proposed by the MathOverflow user Spiderduckpig [17] in the setting of the everywhere unramified tower. Seventh, the finite data are optimized against the sharpened master inequality: we found T , the selected set S , the weights $k(p)$, and the radius R by a computer search combined with an exact optimization (section 7). The fifth and sixth differences are the main new ingredients, and we describe them now.

1.1. Overview of the Dyadic Two Ray Tower. The tower of [14, Lemmas 11 and 12] is everywhere unramified over the real quadratic base field $Q_0 = \mathbb{Q}(\sqrt{D})$, where $D = \prod_{q \in T} q$ for a finite set T of odd primes, while the original construction of [13] used unramified 3-towers over cyclic cubic fields. We allow ramification at the unique dyadic prime $\mathfrak{q}$ of Q_0 , which is unique since $D \equiv 3 \pmod{8}$, but only in a controlled two dimensional direction: every dyadic decomposition group is required to factor through the *two ray quotient*

$$A_{\mathfrak{q}} = \text{Gal} \left(K_{\mathfrak{q}}(\sqrt{5}, \sqrt{2})/K_{\mathfrak{q}} \right) \simeq C_2 \times C_2, \quad K_{\mathfrak{q}} = (Q_0)_{\mathfrak{q}} \simeq \mathbb{Q}_2(\sqrt{3}),$$

whose two rays are the unramified quadratic direction $\sqrt{5}$ and the ramified direction $\sqrt{2}$. The condition keeps the dyadic residue degree at most 2, which the master exponent inequality (Proposition 10) requires, while adjoining $\sqrt{2}$ to the tower.

The ramified direction must be built into the global ramification set from the outset: $Q_0(\sqrt{2})/Q_0$ is ramified at $\mathfrak{q}$, and so it is not a subextension of any everywhere unramified tower, and no bookkeeping of local presentations on the unramified tower can produce it. All relation counting below takes place in the Galois group G_0 of the maximal pro-2 extension of Q_0 unramified outside $\mathfrak{q}$, with no real primes in the allowed ramification set, so that every field in the tower is totally real. Two structural facts make this tractable. First, by a theorem of Schmidt [15], G_0 has cohomological dimension at most 2 and satisfies the exact Euler-Poincaré identity $1 - d(G_0) + r(G_0) = -r_2(Q_0) = 0$, and so G_0 has a minimal presentation with exactly $d(G_0) - 1$ relations. Imposing the finite quotient $A_{\mathfrak{q}}$ at the dyadic decomposition groups costs at most five further relations (Lemma 14), and so the constrained group admits a minimal presentation with at most four more relators than generators before any selected prime conditions are imposed (Proposition 16). Second, a dyadic reciprocity lemma (Lemma 18) shows that rational principal ideles have trivial local Artin image in $A_{\mathfrak{q}}$. By global reciprocity, the two Frobenius

elements above a rational prime p split in Q_0 then agree modulo commutators and squares (Lemma 19), and so once the first square Frobenius relation at p is imposed, the second one has Zassenhaus degree at least 3 (Proposition 20).

Under the weighted Golod-Shafarevich criterion of Vinberg and Koch [18, 9], relations of degree 3 are far cheaper than relations of degree 2. The upshot (Theorem 22) is that the constrained tower is infinite whenever

$$1 - g\tau + (g + |S| + 3)\tau^2 + U_S\tau^3 < 0$$

for some $\tau \in (0, 1)$, where $g = |T|$, $|S|$ is the number of selected primes, and U_S is the number of selected primes split in Q_0 . Split primes, which are the most valuable primes for the numerator, become affordable in large numbers: our certificate uses 114 of them, whereas the certificate of [14] uses none. The idea of cutting class field towers by powers of Frobenius originates in work of Hajir, Maire and Ramakrishna [6].

For our certificate we take T to be the set of the first 38 odd primes, so that $g = 38$ and $\lambda = \sqrt{16D}$, and a selected set S of 314 rational primes with weights $k(p)$, of which 114 split in Q_0 . The weighted Golod-Shafarevich polynomial is

$$P(\tau) = 1 - 38\tau + 355\tau^2 + 114\tau^3, \quad P\left(\frac{6}{115}\right) = -\frac{101}{1520875} < 0,$$

an exact rational computation, and the analytic input is the bound $\log B_{E_{39}}(\lambda) < 0.09$ of Proposition 25. Every discrete assertion in the certificate (primality, splitting types, admissibility witnesses, counts, and the rational negativity of P) is verified by a finite computation in exact integer or rational arithmetic, and a verification script accompanies the source of this paper.

1.2. Preliminaries and Notation. Unit distances are counted as ordered pairs $(u, v) \in U^2$ throughout, and the unordered count differs by exactly a factor of 2. All logarithms are natural. For a number field L , Δ_L denotes its absolute discriminant, $h(L)$ its class number, and $h^+(L)$ its narrow class number. For a CM field K with maximal totally real subfield F we write $h^-(K) = h(K)/h(F)$, which is a positive integer. Rational primes are denoted p or q , and primes of number fields $\mathfrak{p}$, $\mathfrak{q}$, $\mathfrak{P}$. For a pro-2 group G we write $d(G)$ and $r(G)$ for the minimal numbers of generators and relations, and $G_2 \supseteq G_3 \supseteq \cdots$ for the Zassenhaus filtration.

1.3. Outline. Sections 2 through 7 are devoted to the proof of Theorem 1. In section 2 we realize fractional ideals as normalized lattices in $\mathbb{C}^d$ and prove the uniform covolume count, and in section 3 we produce many norm solutions by a pigeonhole argument in the narrow positive relative class group. In section 4 we prove the relative class number bound through the fixed subfield E_{39} , and in section 5 we assemble the covolume count, the pigeonhole, and the class number bound into the master exponent inequality, Theorem 10. In section 6 we construct the dyadic two ray tower and prove the weighted Golod-Shafarevich criterion, Theorem 22. In section 7 we fix the finite data, certify the bound $\log B_{E_{39}}(\lambda) < 0.09$, and evaluate the master inequality, which proves Theorem 1. The Appendix contains the complete selected prime certificate.

2. IDEAL LATTICES AND THE COVOLUME COUNT

In this section we develop the counting side of the argument. We realize an ideal in a CM field as a lattice in $\mathbb{C}^d$, bound the number of lattice points in any ball by volume over covolume, and convert norm solutions $\beta c(\beta) = \alpha$ into unit distance pairs in the plane. In section 3 we then produce an ideal I and a totally positive α admitting many

such norm solutions, by a pigeonhole argument in a narrow class group attached to K/F . The counting scheme is that of [14, Lemmas 2 through 8], and we sharpen it in two places: the packing bound of [14, Lemma 2] is replaced by a uniform covolume count, and the pigeonhole group of [14, Lemma 6] is replaced by a narrow positive group whose order does not grow with the degree. The goal of sections 2 through 5 is the master exponent inequality, Theorem 10, which converts these counts into a unit distance exponent.

Let F be totally real of degree d , and let K/F be a CM quadratic extension. Choose one complex embedding $\sigma_v : K \hookrightarrow \mathbb{C}$ above each real place v of F . Let I be a fractional ideal of K , let $\alpha \in F_{>0}$ be totally positive, and set

$$Q = \#(N_{K/F}(I)/(\alpha)) \quad \text{and} \quad A = Q^{1/(2d)}.$$

We assume throughout this section that

$$(2.1) \quad \sqrt{|\Delta_K|} = \lambda^d.$$

Following [14, Lemma 4], define the *normalized ideal lattice*

$$(2.2) \quad \Lambda = \left\{ \left(\frac{\sigma_v(x)}{\sqrt{\sigma_v(\alpha)}} \right)_{v|\infty} : x \in I \right\} \subset \mathbb{C}^d \simeq \mathbb{R}^{2d},$$

which we equip with the normalized Euclidean norm

$$|z|^2 = \frac{1}{d} \sum_{v|\infty} |z_v|^2.$$

Thus the normalized length of a vector is $1/\sqrt{d}$ times its ordinary Euclidean length in $\mathbb{R}^{2d}$, while volumes and covolumes below are always ordinary $2d$ -dimensional Lebesgue measure, and we take the balls $B_R(w)$ in the normalized norm. In [14, Lemma 4] the same lattice is measured in the sup norm, while the Euclidean normalization is what the covolume count below requires.

The key will be that the number of points of Λ in any ball is controlled by volume over covolume, uniformly in the center. We have the following uniform covolume count:

Proposition 2. *For every $R > 0$ and every $w \in \mathbb{C}^d$,*

$$(2.3) \quad \#(\Lambda \cap B_R(w)) \leq \frac{\text{vol}_{2d}(B_{R+\lambda/(2A)})}{\text{covol}(\Lambda)},$$

and

$$(2.4) \quad \text{covol}(\Lambda) = \frac{(\lambda/2)^d}{A^{2d}}.$$

Consequently, uniformly in w ,

$$(2.5) \quad \frac{1}{2d} \log \#(\Lambda \cap B_R(w)) \leq \log A + \log \left(R + \frac{\lambda}{2A} \right) + \frac{1}{2} \log(2\pi e) - \frac{1}{2} \log \lambda + o(1),$$

where $o(1) \rightarrow 0$ as $d \rightarrow \infty$ for fixed R , λ , and A .

Proof. Let $\Phi : I \hookrightarrow \mathbb{C}^d$ be the Minkowski embedding $x \mapsto (\sigma_v(x))_v$. For a fractional ideal of a CM field,

$$\text{covol}(\Phi(I)) = 2^{-d} \sqrt{|\Delta_K|} N_{K/\mathbb{Q}}(I).$$

Passing from $\Phi(I)$ to (2.2) multiplies the v^{th} complex coordinate by the real scalar $\sigma_v(\alpha)^{-1/2}$, hence multiplies covolume by

$$\prod_{v|\infty} \sigma_v(\alpha)^{-1} = N_{F/\mathbb{Q}}(\alpha)^{-1},$$

and so

$$\text{covol}(\Lambda) = 2^{-d} \sqrt{|\Delta_K|} \frac{N_{K/\mathbb{Q}}(I)}{N_{F/\mathbb{Q}}(\alpha)}.$$

Since

$$Q = \frac{N_{F/\mathbb{Q}}(\alpha)}{N_{K/\mathbb{Q}}(I)} = A^{2d},$$

(2.1) gives (2.4).

To prove (2.3) we bound the covering radius of Λ , which requires a lower bound for the minimum of the dual lattice. Let $\mathfrak{D}_K$ be the different of $K/\mathbb{Q}$. With respect to the ordinary real inner product on $\mathbb{C}^d \simeq \mathbb{R}^{2d}$, the dual lattice is

$$(2.6) \quad \Lambda^* = \left\{ \left(2\sqrt{\sigma_v(\alpha)} \overline{\sigma_v(\eta)} \right)_v : \eta \in \mathfrak{D}_K^{-1} I^{-1} \right\}.$$

Indeed, for $x \in I$ and $\eta \in \mathfrak{D}_K^{-1} I^{-1}$,

$$\sum_v \text{Re} \left(\frac{\sigma_v(x)}{\sqrt{\sigma_v(\alpha)}} 2\sqrt{\sigma_v(\alpha)} \overline{\sigma_v(\eta)} \right) = \text{Tr}_{K/\mathbb{Q}}(x\eta) \in \mathbb{Z},$$

and conversely trace duality identifies the dual of I under the trace pairing with $\mathfrak{D}_K^{-1} I^{-1}$, so (2.6) is exactly the dual lattice.

For non-zero $\eta \in \mathfrak{D}_K^{-1} I^{-1}$, the ideal $(\eta)\mathfrak{D}_K I$ is integral, so

$$|N_{K/\mathbb{Q}}(\eta)| |\Delta_K| N_{K/\mathbb{Q}}(I) \geq 1.$$

For the corresponding vector $y \in \Lambda^*$ in (2.6),

$$\begin{aligned} \prod_v |y_v|^2 &= 4^d N_{F/\mathbb{Q}}(\alpha) |N_{K/\mathbb{Q}}(\eta)| \\ &\geq 4^d \frac{N_{F/\mathbb{Q}}(\alpha)}{|\Delta_K| N_{K/\mathbb{Q}}(I)} = \left(\frac{2A}{\lambda} \right)^{2d}, \end{aligned}$$

and the inequality of arithmetic and geometric means converts this product bound into the normalized dual minimum bound

$$(2.7) \quad \lambda_1(\Lambda^*) \geq \frac{2A}{\lambda}.$$

We will make use of Banaszczyk's transference theorem [2, Theorem 2.2], which states that $\mu(L)\lambda_1(L^*) \leq n/2$ for every lattice L of rank n , where μ is the covering radius and both quantities are measured in the ordinary Euclidean norm. Here $n = 2d$ and ordinary length is $\sqrt{d}$ times normalized length, so that in normalized units the inequality reads

$$\mu(\Lambda)\lambda_1(\Lambda^*) \leq 1,$$

and so by (2.7)

$$(2.8) \quad \mu(\Lambda) \leq \frac{\lambda}{2A}.$$

Now tile $\mathbb{C}^d$ by Voronoi cells of Λ . Each cell has volume $\text{covol}(\Lambda)$ and, by (2.8), is contained in the normalized ball of radius $\lambda/(2A)$ around its lattice point. The cells

centered at the points of $\Lambda \cap B_R(w)$ are disjoint and lie in $B_{R+\lambda/(2A)}(w)$, which proves (2.3). Finally, a ball of normalized radius r has ordinary radius $\sqrt{d}r$, so

$$\text{vol}_{2d}(B_r) = \frac{\pi^d (dr^2)^d}{d!} = \exp(o(d)) (\pi e r^2)^d$$

by Stirling's formula, which together with (2.4) gives (2.5). $\square$

The following lemma converts solutions of $\beta c(\beta) = \alpha$ in I into unit distance pairs in the plane. It plays the role of Lemmas 2 and 5 of [14], with the overlap factor $(1 - \frac{1}{R})^{2d}$ there sharpened to the exact two ball overlap $\Theta_{2d}(R)$:

Proposition 3. *Let M be the number of elements $\beta \in I$ satisfying $\beta c(\beta) = \alpha$. For every fixed $R > 1/2$ there is a finite set $U \subset \mathbb{R}^2$ such that*

$$(2.9) \quad \frac{1}{2d} \log \#U \leq \log A + \log \left(R + \frac{\lambda}{2A} \right) + \frac{1}{2} \log(2\pi e) - \frac{1}{2} \log \lambda + o(1)$$

and

$$(2.10) \quad \frac{\# \{(u, v) \in U^2 : |u - v| = 1\}}{\#U} \geq M \Theta_{2d}(R),$$

where $\Theta_{2d}(R)$ is the volume of the intersection of two $2d$ -dimensional balls of normalized radius R whose centers are at normalized distance 1, divided by the volume of one such ball. In addition,

$$(2.11) \quad \lim_{d \rightarrow \infty} \frac{1}{2d} \log \Theta_{2d}(R) = \frac{1}{2} \log \left(1 - \frac{1}{4R^2} \right).$$

Proof. If $\beta c(\beta) = \alpha$, then $|\sigma_v(\beta)|^2 = \sigma_v(\alpha)$ for every real place v of F , so the corresponding lattice vector $t_\beta \in \Lambda$ has normalized length 1, and its projection to any fixed complex coordinate is a unit vector of $\mathbb{C} \simeq \mathbb{R}^2$ after the same normalization. The projection $\Lambda \rightarrow \mathbb{C}$ is injective since it is induced by a field embedding $K \hookrightarrow \mathbb{C}$.

Now average over translates. Choose w uniformly at random in a fundamental domain of Λ and let $U_w \subset \mathbb{R}^2$ be the projection of $\Lambda \cap B_R(w)$. For each solution vector t_β , the expected number of $x \in \Lambda$ with both x and $x + t_\beta$ in $B_R(w)$ is

$$\frac{\text{vol}(B_R \cap (B_R - t_\beta))}{\text{covol}(\Lambda)} = \Theta_{2d}(R) \frac{\text{vol}(B_R)}{\text{covol}(\Lambda)},$$

while the expected number of points of $\Lambda \cap B_R(w)$ is exactly $\text{vol}(B_R)/\text{covol}(\Lambda)$. Each such pair $(x, x + t_\beta)$ projects to an ordered pair of points of U_w at distance 1. Summing over the M solutions and comparing expectations, we see that some translate w satisfies (2.10), and Proposition 2 bounds $\#U_w$ uniformly in w , so that (2.9) holds for the same translate. The limit (2.11) is the standard large dimension asymptotic for the relative volume of the intersection of two equal balls at center distance 1, since the intersection contains and is contained in caps whose relative volume has exponential rate $\frac{1}{2} \log(1 - 1/(4R^2))$. This then completes the proof of Proposition 3. $\square$

Note that the count (2.3) improves the packing bound $\#U \leq (2RA + 1)^{2d}$ of [14, Lemma 2] in two ways. First, the covolume $(\lambda/2)^d/A^{2d}$ is visible in the denominator, and contributes the term $-\frac{1}{2} \log \lambda$ to (2.5). Second, the additive thickening $\lambda/(2A)$ is negligible for our certificate, where $\log(\lambda/2A) \approx -1017$, so that the count is effectively volume over covolume. This is what makes small radii R usable in Proposition 3.

3. THE NARROW POSITIVE PIGEONHOLE

The count of Proposition 3 requires an ideal I and a totally positive α admitting many solutions of $\beta c(\beta) = \alpha$, and in this section we produce such a pair. The pigeonhole will take place in the following group, which records ideals together with a chosen totally positive generator of the relative norm. Let

$$G_{K/F}^+ = \{(J, u) : J \text{ a fractional ideal of } K, N_{K/F}J = (u), u \gg 0\} / \sim,$$

where

$$(J, u) \sim ((\gamma)J, \gamma c(\gamma)u) \quad \text{for } \gamma \in K^\times.$$

The group $G_{K/F}^+$ may be viewed as a narrow class group of the norm one torus attached to K/F . The corresponding group G_K of [14, Lemma 6] omits the positivity condition on u , and has order as large as $2^{d+1}h^-(K)$. Imposing total positivity gives a bound independent of the degree:

Lemma 4. *If K/F is quadratic and unramified at every finite prime, then*

$$\#G_{K/F}^+ \leq 2h^-(K), \quad \text{where } h^-(K) = \frac{h(K)}{h(F)}.$$

Proof. There is an exact sequence

$$\mathcal{O}_K^\times \longrightarrow \mathcal{O}_F^{\times,+} \longrightarrow G_{K/F}^+ \longrightarrow \text{Cl}(K) \longrightarrow \text{Cl}^+(F),$$

in which the first map is the relative norm $\gamma \mapsto \gamma c(\gamma)$, the second sends a totally positive unit u to the class of $(\mathcal{O}_K, u)$, the third sends (J, u) to the class of J , and the fourth is the relative norm of ideals. Exactness is immediate from the definitions. Indeed, (J, u) maps to zero in $\text{Cl}(K)$ exactly when $J = (\gamma)$ is principal, in which case $(J, u) \sim (\mathcal{O}_K, \gamma^{-1}c(\gamma)^{-1}u)$ comes from the totally positive unit $\gamma^{-1}c(\gamma)^{-1}u$. Similarly, the classes of ideals J with $N_{K/F}J$ principal with a totally positive generator are precisely those mapping to zero in $\text{Cl}^+(F)$.

It follows that

$$\#G_{K/F}^+ \leq \#(\mathcal{O}_F^{\times,+}/N_{K/F}\mathcal{O}_K^\times \cap \mathcal{O}_F^{\times,+}) \cdot \#\ker(\text{Cl}(K) \rightarrow \text{Cl}^+(F)).$$

We first bound the unit factor. Since $N_{K/F}(\gamma) = \gamma^2$ for $\gamma \in \mathcal{O}_F^\times$, we have that $(\mathcal{O}_F^\times)^2 \subseteq N_{K/F}\mathcal{O}_K^\times$, and so the factor is at most $\#(\mathcal{O}_F^{\times,+}/(\mathcal{O}_F^\times)^2)$. Since F is totally real, $\mathcal{O}_F^\times \simeq \{\pm 1\} \times \mathbb{Z}^{d-1}$, hence $\#(\mathcal{O}_F^\times/(\mathcal{O}_F^\times)^2) = 2^d$, and the narrow class number formula $h^+(F)/h(F) = 2^d/[\mathcal{O}_F^\times : \mathcal{O}_F^{\times,+}]$ gives

$$\#(\mathcal{O}_F^{\times,+}/(\mathcal{O}_F^\times)^2) = \frac{2^d}{[\mathcal{O}_F^\times : \mathcal{O}_F^{\times,+}]} = \frac{h^+(F)}{h(F)}.$$

We next bound the class group factor. Since K is totally imaginary, we have $\text{Cl}^+(K) = \text{Cl}(K)$, and since K/F is quadratic, unramified at all finite primes, and ramified at all infinite places, K is contained in the narrow Hilbert class field of F . By class field theory the cokernel of the norm map $\text{Cl}(K) \rightarrow \text{Cl}^+(F)$ then has order at most $[K : F] = 2$, so

$$\#\ker(\text{Cl}(K) \rightarrow \text{Cl}^+(F)) = \frac{h(K)}{\#N_{K/F}\text{Cl}(K)} \leq \frac{2h(K)}{h^+(F)}.$$

Multiplying the two factors gives $\#G_{K/F}^+ \leq 2h(K)/h(F)$, which proves the lemma. $\square$

Armed with Lemma 4, we are now ready to run the pigeonhole argument of [14, Lemma 7] in the smaller group:

Proposition 5. *Let S_F be a finite set of prime ideals of F , all split in K/F , and let $k : S_F \rightarrow \mathbb{Z}_{\geq 1}$. Then there exist a fractional ideal I of K and a totally positive $\alpha \in N_{K/F}(I)$ such that*

$$\#\{\beta \in I : \beta c(\beta) = \alpha\} \geq \frac{\prod_{\mathfrak{p} \in S_F} (k(\mathfrak{p}) + 1)}{h^-(K)}$$

and

$$\#(N_{K/F}(I)/(\alpha)) = \prod_{\mathfrak{p} \in S_F} \#(\mathcal{O}_F/\mathfrak{p})^{k(\mathfrak{p})}.$$

Proof. For each $\mathfrak{p} \in S_F$, write $\mathfrak{p}\mathcal{O}_K = \mathfrak{P}_{\mathfrak{p}}c(\mathfrak{P}_{\mathfrak{p}})$ and consider the set L of ideals

$$J = \prod_{\mathfrak{p} \in S_F} \mathfrak{P}_{\mathfrak{p}}^{j_{\mathfrak{p}}} c(\mathfrak{P}_{\mathfrak{p}})^{k(\mathfrak{p})-j_{\mathfrak{p}}} \quad \text{with} \quad 0 \leq j_{\mathfrak{p}} \leq k(\mathfrak{p}).$$

Every $J \in L$ has relative norm $\prod_{\mathfrak{p}} \mathfrak{p}^{k(\mathfrak{p})}$, and $\#L = \prod_{\mathfrak{p}} (k(\mathfrak{p}) + 1)$. Fix $J_0 \in L$. For $J \in L$ the ideal JJ_0^{-1} has trivial relative norm, so $(JJ_0^{-1}, 1)$ defines a class in $G_{K/F}^+$. By Lemma 4, some fiber of the map $L \rightarrow G_{K/F}^+$, $J \mapsto [(JJ_0^{-1}, 1)]$, has size at least $\#L/(2h^-(K))$.

Fix such a fiber, with class represented by (J_m, u) , $u \gg 0$. For every J in the fiber there is $\beta_J \in K^\times$ with

$$(\beta_J)J_m = JJ_0^{-1} \quad \text{and} \quad u\beta_J c(\beta_J) = 1.$$

Take $I = J_0^{-1}J_m^{-1}$ and $\alpha = u^{-1}$, which is totally positive. Then $\beta_J \in I$ and $\beta_J c(\beta_J) = \alpha$, and we have that $N_{K/F}(I) = (\alpha) \prod_{\mathfrak{p}} \mathfrak{p}^{-k(\mathfrak{p})}$, which gives the stated index $\#(N_{K/F}(I)/(\alpha))$. Distinct J in the fiber determine distinct ideals $(\beta_J)J_m$, hence distinct solutions β_J . Since $-\beta_J$ is a further solution generating the same ideal, the solutions come in distinct pairs $\pm\beta_J$. The total count is thus at least

$$2 \cdot \frac{\#L}{2h^-(K)} = \frac{\prod_{\mathfrak{p}} (k(\mathfrak{p}) + 1)}{h^-(K)}. \quad \square$$

Specializing to rational primes as in [14, Lemma 8], we have the following corollary:

Corollary 6. *Assume $K/\mathbb{Q}$ is Galois with totally real subfield F of degree d . Let S be a finite set of rational primes and let $k : S \rightarrow \mathbb{Z}_{\geq 1}$. Suppose every prime of F above every $p \in S$ splits in K/F . Let $e(p)$ be the ramification index of p in $F/\mathbb{Q}$ and let $f(p)$ be an upper bound for the inertia degree of p in $F/\mathbb{Q}$. Then I, α may be chosen so that*

$$M := \#\{\beta \in I : \beta c(\beta) = \alpha\} \geq \frac{\prod_{p \in S} (k(p) + 1)^{d/(e(p)f(p))}}{h^-(K)}$$

and

$$\#(N_{K/F}(I)/(\alpha)) = \prod_{p \in S} p^{k(p)d/e(p)}.$$

Proof. Apply Proposition 5 with S_F the set of all primes of F above the primes of S , taking the weight of every prime above p to be $k(p)$. Since $F/\mathbb{Q}$ is Galois, there are exactly $d/(e(p)f_p)$ primes above p , where $f_p \leq f(p)$ is the true inertia degree, and each has residue field of size p^{f_p} . Hence

$$\prod_{\mathfrak{p}|p} (k(\mathfrak{p}) + 1) = (k(p) + 1)^{d/(e(p)f_p)} \geq (k(p) + 1)^{d/(e(p)f(p))},$$

while the norm index is $\prod_{\mathfrak{p}|p} p^{f_{\mathfrak{p}}k(p)} = p^{k(p)d/e(p)}$, independent of f_p , and the result follows. $\square$

4. THE RELATIVE CLASS NUMBER BOUND

In this section we prove an upper bound for the relative class number $h^-(K)$, which the pigeonhole of section 3 loses as a denominator. In section 5 we will then assemble the covolume count, the pigeonhole, and the class number bound into the master exponent inequality. Louboutin's method [10] bounds $L(1, \chi_{K/F})$ by an expression involving ζ_F near $s = 1$, and bounding ζ_F crudely costs a factor that grows like $\log \lambda$ per unit of degree. The towers constructed in section 6 will all contain one fixed multiquadratic field E however, and the degree normalized Dedekind zeta function of this subfield majorizes that of every field F in the tower (Lemma 7). Since the splitting of rational primes in a multiquadratic field is completely explicit, we are able to certify an upper bound for ζ_E near $s = 1$ numerically once and for all in section 7.1, and the loss of size roughly $\log \lambda$ is replaced by a small explicit constant.

Let $E \subseteq F$ be a fixed subfield. For $\lambda > 1$ define

$$(4.1) \quad B_E(\lambda) = \inf_{x>0} \lambda^{x/2} \pi^{-x/2} \Gamma(1+x/2) \zeta_E(1+x)^{1/[E:\mathbb{Q}]}.$$

We first show that the degree normalized zeta function of a field is dominated by that of any subfield:

Lemma 7. *If $E \subseteq F$ and $\sigma > 1$, then*

$$\zeta_F(\sigma)^{1/[F:\mathbb{Q}]} \leq \zeta_E(\sigma)^{1/[E:\mathbb{Q}]}.$$

Proof. It suffices to compare, for each prime $\mathfrak{p}$ of E , the local factors above $\mathfrak{p}$. Put $q = N_{E/\mathbb{Q}}\mathfrak{p}$ and $m = [F:E]$. If the primes of F above $\mathfrak{p}$ have residue degrees $f_1, \dots, f_r$ over $\mathfrak{p}$, then $r \leq m$, and the degree normalized local contribution to $\log \zeta_F$ is

$$\frac{1}{m} \sum_{i=1}^r -\log(1 - q^{-f_i\sigma}) \leq \frac{r}{m} (-\log(1 - q^{-\sigma})) \leq -\log(1 - q^{-\sigma}),$$

since $f_i \geq 1$ and $-\log(1 - t)$ is increasing for $t \in (0, 1)$. The right side is the degree normalized local contribution to $\log \zeta_E$, and so summing over $\mathfrak{p}$ and exponentiating, the result follows. $\square$

With Lemma 7 in hand, we are now ready to prove our version of Louboutin's bound. The corresponding bound is Lemma 9 of [14], which reads $h^-(K) \leq 8\lambda^2 \left(\frac{e}{4\pi} \sqrt{\lambda} \log \lambda \right)^d$, and the effect of the fixed subfield is to replace the factor $\frac{e}{2} \log \lambda$ by $B_E(\lambda)$:

Proposition 8. *Let K/F be a CM quadratic extension with associated quadratic character $\chi_{K/F}$, and put*

$$\lambda = (\Delta_K/\Delta_F)^{1/d} \quad \text{and} \quad d = [F:\mathbb{Q}].$$

If $E \subseteq F$, then

$$L(1, \chi_{K/F}) \leq 2 B_E(\lambda)^d.$$

Consequently

$$(4.2) \quad h^-(K) \leq C_\lambda \left(\frac{\sqrt{\lambda} B_E(\lambda)}{2\pi} \right)^d,$$

where C_λ depends only on λ and not on d , and we may take $C_\lambda = 8\lambda^2$.

Proof. Louboutin's positive kernel argument [10, Theorem 2] bounds $L(1, \chi_{K/F})$ by the value of the completed L -function at $1+x$, for any $x > 0$, after replacing the Dirichlet coefficients by their absolute values, and those absolute values are dominated by the coefficients of ζ_F . Since K is totally imaginary and F totally real, every one of the d archimedean places contributes the odd gamma factor, and we obtain, for every $x > 0$,

$$L(1, \chi_{K/F}) \leq 2 \left(\lambda^{x/2} \pi^{-x/2} \Gamma(1+x/2) \right)^d \zeta_F(1+x).$$

Applying Lemma 7 to bound $\zeta_F(1+x) \leq \zeta_E(1+x)^{d/[E:\mathbb{Q}]}$ and taking the infimum over $x > 0$ gives the first assertion.

For the second assertion, the relative class number formula for CM fields reads

$$h^-(K) = Q_K w_K \left(\frac{\sqrt{\lambda}}{2\pi} \right)^d L(1, \chi_{K/F}),$$

where $Q_K \in \{1, 2\}$ is the Hasse unit index and w_K is the number of roots of unity in K . We claim that $w_K \leq 2\lambda^2$. To see this, we make use of the fact that the field K contains $\mathbb{Q}(\mu_{w_K})$, that root discriminants do not decrease in extensions of number fields, and that

$$\text{rd}(\mathbb{Q}(\mu_m)) = \frac{m}{\prod_{p|m} p^{1/(p-1)}} \geq \sqrt{m/2}$$

for all $m \geq 3$ (see [19, Proposition 2.7] for the discriminant of cyclotomic fields). Since $|\Delta_K| = \Delta_F^2 N_{F/\mathbb{Q}}(\mathfrak{d}_{K/F}) \geq \Delta_F^2$, we have that

$$\text{rd}(K) = |\Delta_K|^{1/2d} \leq (\Delta_K/\Delta_F)^{1/d} = \lambda,$$

and so $w_K \leq 2\text{rd}(K)^2 \leq 2\lambda^2$, which proves the claim. The bounded factors $Q_K w_K \leq 4\lambda^2$ together with the factor 2 from the first assertion are absorbed into $C_\lambda = 8\lambda^2$, and the result follows. $\square$

Before assembling the exponent, we record the bookkeeping of the narrow positive factor:

Remark 9. On the $(2d)^{-1} \log$ scale, (4.2) and Proposition 5 together contribute

$$\frac{1}{2} \log(2\pi) - \frac{1}{4} \log \lambda - \frac{1}{2} \log B_E(\lambda)$$

to the numerator of the master inequality. Had we used the non-narrow pigeonhole with denominator $2^d h^-(K)$, the constant $\frac{1}{2} \log(2\pi)$ would read $\frac{1}{2} \log \pi$. The narrow positive saving of Lemma 4 is thus already incorporated exactly once, and adding a further $\frac{1}{2} \log 2$ anywhere in the accounting would count it twice.

5. THE MASTER EXPONENT INEQUALITY

Armed with the covolume count of section 2, the narrow positive pigeonhole of section 3, and the class number bound of Proposition 8, we are now ready to prove the master exponent inequality, which turns any family of CM fields of growing degree with prescribed splitting, ramification, and root discriminant data into a unit distance exponent expressed as a ratio of two explicit logarithmic sums:

Theorem 10. *Let the subfield E and the real number $\lambda > 1$ be fixed. Suppose there are Galois CM fields K of arbitrarily large degree with totally real subfields F , $d = [F:\mathbb{Q}]$, satisfying the following four conditions. First, $E \subseteq F$. Second, $(\Delta_K/\Delta_F)^{1/d} = \lambda$ and $\sqrt{|\Delta_K|} = \lambda^d$. Third, every prime of F above every $p \in S$ splits in K/F . Fourth, $e(p)$*

is the ramification index of p in $F/\mathbb{Q}$, and $f(p)$ is an upper bound for the inertia degree of p in $F/\mathbb{Q}$. Let $k : S \rightarrow \mathbb{Z}_{\geq 1}$ and put

$$A = \prod_{p \in S} p^{k(p)/(2e(p))}.$$

For $R > 1/2$ define

$$(5.1) \quad N(R) = \frac{1}{2} \log \left(1 - \frac{1}{4R^2} \right) + \sum_{p \in S} \frac{\log(k(p) + 1)}{2e(p)f(p)} + \frac{1}{2} \log(2\pi) - \frac{1}{4} \log \lambda - \frac{1}{2} \log B_E(\lambda)$$

and

$$(5.2) \quad M(R) = \log A + \log \left(R + \frac{\lambda}{2A} \right) + \frac{1}{2} \log(2\pi e) - \frac{1}{2} \log \lambda.$$

If $M(R) > 0$, then for every $\varepsilon > 0$ there are arbitrarily large finite sets $U \subset \mathbb{R}^2$ with

$$\# \{ (u, v) \in U^2 : |u - v| = 1 \} \geq (\#U)^{1+N(R)/M(R)-\varepsilon}.$$

In particular every exponent below $1 + N(R)/M(R)$ is achieved by arbitrarily large finite planar point sets.

Theorem 10 is our version of the master criterion of [14, Proposition 10], and the differences are exactly the sharpenings assembled above. In the numerator, the overlap term $\log(1 - \frac{1}{R})$ of [14] becomes the exact two ball exponent $\frac{1}{2} \log(1 - \frac{1}{4R^2})$ by Proposition 3, while the terms $\frac{1}{2} \log(2\pi/e) - \frac{1}{2} \log \log \lambda$ there, which record the pigeonhole and class number losses, become $\frac{1}{2} \log(2\pi) - \frac{1}{2} \log B_E(\lambda)$ by Lemma 4 and Proposition 8. In the denominator, the packing bound $\log(2RA + 1)$ of [14] becomes the covolume count of Proposition 2.

Proof. If $N(R) \leq 0$ then $1 + N(R)/M(R) \leq 1$, and the conclusion already holds for n collinear points spaced at unit distance, which span $2(n - 1)$ ordered unit distance pairs, and so we may assume $N(R) > 0$.

Fix K and F in the family with d large. Our goal is to combine Corollary 6, which produces many solutions of $\beta c(\beta) = \alpha$ in an ideal, Proposition 8, which bounds the class number loss, and Proposition 3, which converts these solutions into unit distance pairs in the plane. Corollary 6 supplies I and α with $Q = \#(N_{K/F}(I)/(\alpha)) = A^{2d}$ and

$$\frac{1}{2d} \log M_I \geq \sum_{p \in S} \frac{\log(k(p) + 1)}{2e(p)f(p)} - \frac{1}{2d} \log h^-(K),$$

where $M_I = \# \{ \beta \in I : \beta c(\beta) = \alpha \}$. Applying Proposition 8 to bound the class number term, we obtain

$$\frac{1}{2d} \log M_I \geq \sum_{p \in S} \frac{\log(k(p) + 1)}{2e(p)f(p)} + \frac{1}{2} \log(2\pi) - \frac{1}{4} \log \lambda - \frac{1}{2} \log B_E(\lambda) + o(1),$$

where the $o(1)$ absorbs $(2d)^{-1} \log C_\lambda \rightarrow 0$. Proposition 3, applied with this I and α , produces a set U with

$$\frac{1}{2d} \log \frac{\# \{ (u, v) \in U^2 : |u - v| = 1 \}}{\#U} \geq \frac{1}{2d} \log M_I + \frac{1}{2} \log \left(1 - \frac{1}{4R^2} \right) + o(1) \geq N(R) + o(1)$$

and $\frac{1}{2d} \log \#U \leq M(R) + o(1)$. Combining these two bounds, since $M(R) > 0$ we conclude that

$$\# \{ (u, v) \in U^2 : |u - v| = 1 \} \geq (\#U)^{1+N(R)/M(R)+o(1)},$$

where $o(1) \rightarrow 0$ as $d \rightarrow \infty$ along the family. Note that $\#U \rightarrow \infty$, since the count of unit distance pairs, and hence $\#U$ itself, is at least $e^{2d(N(R)+o(1))}$ with $N(R) > 0$. Taking d large enough that the $o(1)$ is below ε , the result follows. $\square$

6. THE DYADIC TWO RAY TOWER

In this section we construct the fields required by Theorem 10. The everywhere unramified counterpart of this construction is Lemmas 11 and 12 of [14], and the ramified two ray direction at the dyadic prime is what is new. The argument runs as follows. Over the real quadratic field $k = Q_0 = \mathbb{Q}(\sqrt{D})$ we consider the maximal pro-2 extension unramified outside the unique dyadic prime $\mathfrak{q}$, with no real primes in the allowed ramification set. By a theorem of Schmidt its Galois group G_0 satisfies an exact Euler-Poincaré identity, and so it has a minimal presentation with exactly $d(G_0) - 1$ relations. We will then impose a finite two dimensional condition at $\mathfrak{q}$, the *two ray* condition, which keeps the dyadic residue degree bounded while retaining a ramified $\sqrt{2}$ direction, and this will cost at most five relations. The resulting group still surjects onto an elementary abelian quotient of rank $|T|$, which bounds its generator rank from below. A dyadic reciprocity lemma then shows that for each selected rational prime split in k , the two square Frobenius conditions cost one relation of Zassenhaus degree 2 and one of degree at least 3. Feeding these counts into the weighted Golod-Shafarevich criterion of Vinberg and Koch yields the infinitude criterion (6.4), and we extract the tower fields at the end of the section.

Let T be a finite set of odd primes and put

$$D = \prod_{q \in T} q \quad \text{and} \quad k = Q_0 = \mathbb{Q}(\sqrt{D}).$$

Assume throughout this section that

$$D \equiv 3 \pmod{8}$$

and that the number of primes $q \in T$ with $q \equiv 3 \pmod{4}$ is odd. Then k is real quadratic, 2 is ramified in k , there is a unique dyadic prime $\mathfrak{q}$ of k , and

$$k_{\mathfrak{q}} \simeq \mathbb{Q}_2(\sqrt{3}).$$

Let S be a finite set of rational *selected* primes with $2 \in S$. For every odd $p \in S$ assume, as in [14, Lemma 12], the admissibility condition

$$(6.1) \quad p \equiv 1 \pmod{4} \quad \text{or} \quad p \text{ is inert in } \mathbb{Q}(\sqrt{q}) \text{ for some } q \in T,$$

which will guarantee in Proposition 23 that p splits in K/F , where F ranges over the tower fields constructed there and $K = F(i)$. Define

$$U_S = \# \{p \in S : p \neq 2 \text{ and } p \text{ splits in } k\}.$$

Write $K_{\mathfrak{q}} = k_{\mathfrak{q}}$ (we reserve the letter K for the CM fields, which do not appear until Proposition 23). Since $K_{\mathfrak{q}} \simeq \mathbb{Q}_2(\sqrt{3})$, its unramified quadratic extension is $K_{\mathfrak{q}}(\sqrt{5})$. The *dyadic two ray quotient* is

$$(6.2) \quad A_{\mathfrak{q}} = \text{Gal} \left(K_{\mathfrak{q}}(\sqrt{5}, \sqrt{2}) / K_{\mathfrak{q}} \right) = \text{Gal} \left(k_{\mathfrak{q}}^{\text{ur}, (2)} k_{\mathfrak{q}}(\sqrt{2}) / k_{\mathfrak{q}} \right) \simeq C_2^{\text{ur}} \times C_2^{\text{ram}},$$

where $k_{\mathfrak{q}}^{\text{ur}, (2)}$ denotes the unramified quadratic extension, and not the full unramified pro-2 extension. Its two rays are the unramified direction $\sqrt{5}$ and the ramified direction $\sqrt{2}$. Requiring dyadic decomposition groups to factor through this finite quotient keeps

the dyadic residue degree at most 2, as the master inequality requires, while adjoining the ramified $\sqrt{2}$ direction to the tower.

6.1. The Global Base Group. We first set up the presentation bookkeeping promised above. For a finitely generated pro-2 group G , write

$$d(G) = \dim_{\mathbb{F}_2} H^1(G, \mathbb{F}_2) \quad \text{and} \quad r(G) = \dim_{\mathbb{F}_2} H^2(G, \mathbb{F}_2).$$

In a minimal pro-2 presentation $1 \rightarrow R \rightarrow F_0 \rightarrow G \rightarrow 1$, with F_0 free pro-2, $d(G)$ is the number of generators and $r(G)$ the minimal number of relators normally generating R . We write $G = G_1 \supseteq G_2 \supseteq G_3 \supseteq \cdots$ for the Zassenhaus filtration of G (the mod-2 dimension subgroups), so that $G_2 = \Phi(G)$ is the Frattini subgroup, and we say a relator has *Zassenhaus degree* at least m if it lies in $F_{0,m}$, the m^{th} filtration step of the free group. Minimal relators always lie in $F_{0,2}$, that is, they have degree at least 2.

Let

$$G_0 = G_{\{\mathfrak{q}\}}(k)(2)$$

be the Galois group of the maximal pro-2 extension of k unramified outside the finite prime $\mathfrak{q}$, with no real prime included in the ramification set. Since $\mathbb{C}/\mathbb{R}$ counts as ramified at a real place, excluding the real places forces every finite subextension of this tower to be totally real. The relation count for G_0 will come from the following theorem:

Theorem 11. (*Schmidt [15]*) *Let L be a number field and let S_L be a finite set of primes of L containing all primes above 2 and no real primes. Then*

$$\text{cd } G_{S_L}(L)(2) \leq 2.$$

The cohomology groups $H^i(G_{S_L}(L)(2), \mathbb{F}_2)$ are finite for $i = 0, 1, 2$, and

$$\sum_{i=0}^2 (-1)^i \dim_{\mathbb{F}_2} H^i(G_{S_L}(L)(2), \mathbb{F}_2) = -r_2(L),$$

where $r_2(L)$ is the number of complex places of L .

We refer to [15] for the proof and to [12, Chapters VIII and X] for the general theory of Galois groups with restricted ramification. Applied to G_0 , Schmidt's theorem gives an exact relation count:

Lemma 12. *The group G_0 has a minimal pro-2 presentation with $d_0 = d(G_0)$ generators and exactly*

$$r(G_0) = d_0 - 1$$

relations, all of Zassenhaus degree at least 2.

Proof. The set $\{\mathfrak{q}\}$ contains all primes of k above 2, since 2 ramifies in k and $\mathfrak{q}$ is the unique dyadic prime, and it contains no real prime, and so the conditions of Theorem 11 are satisfied. As k is real quadratic, $r_2(k) = 0$, and Theorem 11 gives

$$1 - d(G_0) + r(G_0) = 0,$$

because $H^0(G_0, \mathbb{F}_2)$ has dimension 1. Hence $r(G_0) = d(G_0) - 1$, and minimal relators lie in the Frattini subgroup of the free group, so they have Zassenhaus degree at least 2. $\square$

The next elementary lemma, the bookkeeping device used throughout, shows that killing an element of a pro-2 group either adds one relation or removes one generator:

Lemma 13. *Let G have a minimal pro-2 presentation on d generators with r relators, all of Zassenhaus degree at least 2. Let $x \in G$ and*

$$G' = G / \langle\langle x \rangle\rangle.$$

If $x \in \Phi(G)$, then G' has a minimal presentation on d generators with at most $r + 1$ relators, all of degree at least 2. If $x \notin \Phi(G)$, then G' has a minimal presentation on $d - 1$ generators with at most r relators, all of degree at least 2.

Proof. Let $1 \rightarrow R \rightarrow F \rightarrow G \rightarrow 1$ be the given minimal presentation, so that F is free pro-2 on d generators and $R \subseteq \Phi(F)$. Lift x to $w \in F$.

If $x \in \Phi(G)$, then w may be chosen in $\Phi(F)$, because the preimage of $\Phi(G)$ in F is $\Phi(F)R = \Phi(F)$. Adding w to the relators presents G' on the same generators with $r + 1$ relators of degree at least 2. If $x \notin \Phi(G)$, then $w \notin \Phi(F)$, so w extends to a free pro-2 basis of F . Killing w leaves a free pro-2 group F' on $d - 1$ generators, and the images of the old relators lie in the image of $\Phi(F)$, which is $\Phi(F')$. Hence G' is presented on $d - 1$ generators by at most r relators of degree at least 2. $\square$

6.2. The Finite Dyadic Local Quotient. Let

$$G_{\text{loc}} = G_{K_q}(2) \quad \text{and} \quad N_{\text{loc}} = \ker(G_{\text{loc}} \twoheadrightarrow A_q),$$

where $G_{K_q}(2)$ is the Galois group of the maximal pro-2 extension of K_q . We first show that the two ray condition costs at most five local relations:

Lemma 14. *The closed normal subgroup N_{loc} is normally generated in G_{loc} by at most five elements.*

Proof. Local Kummer theory gives

$$d(G_{\text{loc}}) = \dim_{\mathbb{F}_2} K_q^\times / K_q^{\times 2} = [K_q : \mathbb{Q}_2] + 2 = 4.$$

Choose elements a, b of G_{loc} whose images generate A_q , and extend their classes modulo $\Phi(G_{\text{loc}})$ to a basis of $G_{\text{loc}}/\Phi(G_{\text{loc}})$ by two further elements z_1, z_2 lying in N_{loc} , so that a, b, z_1, z_2 generate G_{loc} . The quotient $A_q \simeq C_2 \times C_2$ has presentation

$$\langle \sigma, \tau \mid \sigma^2, \tau^2, [\sigma, \tau] \rangle.$$

Killing the five elements

$$z_1, z_2, a^2, b^2, [a, b]$$

normally in G_{loc} produces a quotient generated by the images of a, b subject to the defining relations of $C_2 \times C_2$, which still surjects onto A_q . Hence the quotient is exactly A_q , and the five elements normally generate N_{loc} . $\square$

Remark 15. The number five is exactly what local cohomology predicts. The five term exact sequence of

$$1 \longrightarrow N_{\text{loc}} \longrightarrow G_{\text{loc}} \longrightarrow A_q \longrightarrow 1$$

gives

$$\dim_{\mathbb{F}_2} H^1(N_{\text{loc}}, \mathbb{F}_2)^{G_{\text{loc}}} = (4 - 2) + \dim_{\mathbb{F}_2} \ker(H^2(A_q, \mathbb{F}_2) \rightarrow H^2(G_{\text{loc}}, \mathbb{F}_2)).$$

The group $H^2(C_2^2, \mathbb{F}_2)$ is three dimensional, spanned by the cup products of the two Kummer characters attached to 5 and 2. These inflate to the classes of the quaternion algebras (5, 5), (5, 2), and (2, 2) over K_q . Since 5 and 2 lie in $\mathbb{Q}_2$ and $[K_q : \mathbb{Q}_2] = 2$, restriction multiplies Brauer invariants by 2 and kills every 2-torsion invariant. Hence all three classes vanish over K_q , giving $2 + 3 = 5$ normal generators.

Choose an embedding of a separable closure of k into a separable closure of $K_{\mathfrak{q}}$. This determines a decomposition homomorphism

$$\iota : G_{\text{loc}} \longrightarrow G_0.$$

Let $w_1, \dots, w_5$ be local normal generators of N_{loc} as in Lemma 14, and set

$$(6.3) \quad H = G_0 / \langle \iota(w_1), \dots, \iota(w_5) \rangle.$$

We then have the following relation count before any selected prime conditions are imposed:

Proposition 16. *The group H is the Galois group of the maximal pro-2 extension of k satisfying the following three conditions: (a) the extension is totally real, (b) it is unramified at every finite prime other than $\mathfrak{q}$, and (c) every dyadic decomposition group has image factoring through $A_{\mathfrak{q}}$. It has a minimal pro-2 presentation with $d = d(H)$ generators and at most*

$$d + 4$$

relations, all of Zassenhaus degree at least 2.

Proof. The first two conditions are built into G_0 : the only finite prime in the allowed ramification set is $\mathfrak{q}$, and no real prime is in the set. Quotienting by the normal closure of $\iota(w_1), \dots, \iota(w_5)$ forces the chosen dyadic decomposition subgroup to factor through $A_{\mathfrak{q}}$. Since all extensions in the tower are Galois over k and the dyadic decomposition subgroups above $\mathfrak{q}$ are conjugate, the normal closure imposes the same condition at every one of them. Conversely, any finite pro-2 extension of k with properties (a), (b), and (c) kills the image of N_{loc} and hence is a quotient of H .

By Lemma 12, G_0 has $d_0 = d(G_0)$ generators and $d_0 - 1$ relators of degree at least 2. We pass from G_0 to H by killing five elements in succession. Say m of the five are non-Frattini in the intermediate quotient at the moment they are killed. By Lemma 13, those m cuts each remove one generator and add no relator, while the other $5 - m$ cuts each add at most one relator of degree at least 2. The final generator rank is $d = d_0 - m$ and the relator count is at most

$$(d_0 - 1) + (5 - m) = d + 4. \quad \square$$

6.3. The Elementary Quotient and Dyadic Reciprocity. Put

$$E_T = k(\sqrt{q} : q \in T, \sqrt{2}).$$

We first check that this elementary abelian quotient survives the two ray condition:

Lemma 17. *The extension E_T/k is a quotient of H and*

$$\text{Gal}(E_T/k) \simeq (\mathbb{Z}/2\mathbb{Z})^{|T|}.$$

Consequently $d(H) \geq g := |T|$.

Proof. The genus part $k(\sqrt{q} : q \in T)/k$ has rank $|T| - 1$: the rational square classes $q \in T$ satisfy exactly one relation over k , namely that their product D is a square in k . The standard genus theory computation shows this extension is unramified at every finite prime and totally real: above a given $q \in T$, the extension is generated over k by the square roots of the primes $q' \in T \setminus \{q\}$, which are units at q (note $\sqrt{q} \in k(\sqrt{q'} : q' \neq q)$ because $\sqrt{D} \in k$). At 2, each $q' \in T$ has local square class 1 or 5 in $K_{\mathfrak{q}}$, and both classes generate unramified directions, so the genus part is unramified at $\mathfrak{q}$ as well.

The class $\sqrt{2}$ is independent of the genus part, because $k(\sqrt{2})/k$ is ramified at $\mathfrak{q}$ while the genus part is unramified at all finite primes. Hence $\text{Gal}(E_T/k)$ is elementary abelian of rank $(|T| - 1) + 1 = |T|$.

What remains to be shown is the dyadic two ray condition (c). In $K_{\mathfrak{q}} = \mathbb{Q}_2(\sqrt{3})$ the class of 3 is a square, so the image of $\mathbb{Q}_2^{\times}/\mathbb{Q}_2^{\times 2}$ in $K_{\mathfrak{q}}^{\times}/K_{\mathfrak{q}}^{\times 2}$ is generated by the classes of 2 and 5. Each odd $q \in T$ has local square class 1 or 5 in $K_{\mathfrak{q}}$, while $\sqrt{2}$ generates the class of 2. It follows that the completion of E_T at any dyadic prime is contained in $K_{\mathfrak{q}}(\sqrt{5}, \sqrt{2})$, that is, the dyadic decomposition group factors through $A_{\mathfrak{q}}$. Hence E_T is a quotient of H , and $d(H) \geq d(\text{Gal}(E_T/k)) = |T|$. $\square$

The following reciprocity lemma is the key arithmetic input behind the split prime saving, and shows that the dyadic ray conditions are invisible to rational ideles:

Lemma 18. *For every $u \in \mathbb{Q}_2^{\times} \subset K_{\mathfrak{q}}^{\times}$, the local Artin image of u in*

$$A_{\mathfrak{q}} = \text{Gal}(K_{\mathfrak{q}}(\sqrt{5}, \sqrt{2})/K_{\mathfrak{q}})$$

is trivial. Equivalently, the Hilbert symbols

$$(u, 5)_{K_{\mathfrak{q}}} = (u, 2)_{K_{\mathfrak{q}}} = 1$$

vanish for all $u \in \mathbb{Q}_2^{\times}$.

Proof. The two characters of $A_{\mathfrak{q}}$ are the Kummer characters of the square classes 5 and 2. By local class field theory, the value of the character attached to $a \in \{5, 2\}$ on $\text{Art}_{\mathfrak{q}}(u)$ is the Hilbert symbol $(a, u)_{K_{\mathfrak{q}}}$. Since both a and u lie in $\mathbb{Q}_2^{\times}$, the quaternion class $(a, u)_{K_{\mathfrak{q}}}$ is the restriction to $K_{\mathfrak{q}}$ of the class $(a, u)_{\mathbb{Q}_2}$, and restriction on Brauer groups multiplies invariants by the degree:

$$\text{inv}_{K_{\mathfrak{q}}}(\text{res}_{K_{\mathfrak{q}}/\mathbb{Q}_2} B) = [K_{\mathfrak{q}} : \mathbb{Q}_2] \text{inv}_{\mathbb{Q}_2}(B) = 2 \text{inv}_{\mathbb{Q}_2}(B).$$

Every 2-torsion Brauer invariant is 0 or $1/2$ in $\mathbb{Q}/\mathbb{Z}$, so multiplication by 2 annihilates it. Thus $(a, u)_{K_{\mathfrak{q}}} = 1$ for $a \in \{5, 2\}$, and both characters of $A_{\mathfrak{q}}$ vanish on $\text{Art}_{\mathfrak{q}}(u)$. $\square$

6.4. The Split Prime Degree Three Relation. Let $p \neq 2$ be a rational prime split in k :

$$p\mathcal{O}_k = \mathfrak{p}\bar{\mathfrak{p}}.$$

Let $x = \text{Frob}_{\mathfrak{p}}$ and $y = \text{Frob}_{\bar{\mathfrak{p}}}$ denote Frobenius elements in the dyadically constrained group $(H, \text{ or any quotient of } H \text{ by selected prime conditions at other primes})$ before the two square Frobenius cuts at p are imposed. Frobenius elements are well defined here because $\mathfrak{p}$ and $\bar{\mathfrak{p}}$ are unramified in the tower. Dyadic reciprocity now gives the following compatibility:

Lemma 19. *The two Frobenius elements have the same image in the mod-2 abelianization:*

$$x \equiv y \pmod{H_2}.$$

Proof. It suffices to test equality against every quadratic character χ of H . Such a character corresponds to a quadratic extension of k that is totally real, unramified at all finite primes other than $\mathfrak{q}$, and whose dyadic local character factors through $A_{\mathfrak{q}}$.

Apply global Artin reciprocity to the principal idele of the positive rational number p . The real local factors are trivial because the corresponding extension is totally real and p is positive in both real embeddings of k . The dyadic local factor is trivial by Lemma 18, since $p \in \mathbb{Q}_2^{\times}$. Every other finite place is unramified for χ and p is a local

unit there, except at $\mathfrak{p}$ and $\bar{\mathfrak{p}}$, where the local factors are $\chi(x)$ and $\chi(y)$. The product formula then reads

$$\chi(x)\chi(y) = 1,$$

and since the values are ± 1 this says $\chi(x) = \chi(y)$ for every quadratic character χ of H . Hence $x \equiv y \pmod{H_2}$. $\square$

It follows that the second square Frobenius relation may be counted in degree at least 3:

Proposition 20. *After imposing the first square Frobenius relation $x^2 = 1$ for a rational prime p split in k , the second square Frobenius relation $y^2 = 1$ can be represented by a relator of Zassenhaus degree at least 3.*

Proof. By Lemma 19, write $y = xh$ with $h \in H_2$. In the quotient in which $x^2 = 1$ has been imposed,

$$y^2 = (xh)^2 = x^2(x^{-1}hx)h = (x^{-1}hx)h.$$

Since $h \in H_2$, the Zassenhaus commutator inclusion $[H_i, H_j] \subseteq H_{i+j}$ gives $x^{-1}hx \equiv h \pmod{H_3}$, so

$$(x^{-1}hx)h \equiv h^2 \pmod{H_3}.$$

For the mod-2 Zassenhaus filtration, $h \in H_2$ implies $h^2 \in H_4 \subseteq H_3$, so $y^2 \in H_3$. If $F_0 \twoheadrightarrow H$ is a minimal free pro-2 presentation, functoriality of the Zassenhaus filtration gives $F_{0,3} \twoheadrightarrow H_3$, so y^2 lifts to a relator lying in $F_{0,3}$, that is, of degree at least 3. $\square$

6.5. The Weighted Golod-Shafarevich Criterion. We will make use of the following weighted refinement of the Golod-Shafarevich theorem [5], due to Vinberg [18] and Koch [9] (see also [12, Chapter III, §9]):

Theorem 21. *(Vinberg, Koch) Let G be a pro-2 group with a presentation on d generators whose relators $\rho_1, \rho_2, \dots$ have Zassenhaus degrees $m_1, m_2, \dots \geq 2$. If there exists $\tau \in (0, 1)$ with*

$$1 - d\tau + \sum_i \tau^{m_i} < 0,$$

then G is infinite.

Starting from H , we impose the selected prime conditions for the primes of S as follows. For $p = 2$ no cut is needed, since the dyadic residue degree bound is already encoded in A_q . If an odd $p \in S$ is inert in k , we kill the Frobenius of the unique prime of k above p . If p is ramified or split in k , we impose the square Frobenius relations at the (one or two) primes of k above p . For split p , the first square Frobenius relation is counted in degree 2 and the second in degree at least 3, by Proposition 20. Killing Frobenius elements or their squares to force bounded residue degrees in a class field tower follows Hajir, Maire and Ramakrishna [6], and the weighted count for split primes, with the second square Frobenius relation placed at degree 3, was proposed by the MathOverflow user Spiderduckpig [17]. We are now ready to prove the main criterion of this section, the two ray analog of [14, Lemma 11], where the unweighted criterion counts both square Frobenius relations of a split prime at the same cost:

Theorem 22. *With the notation above, the pro-2 group obtained after all of the selected prime cuts has a presentation with generator rank $d_{\text{fin}} \geq g = |T|$, at most*

$$d_{\text{fin}} + |S| + 3$$

relators of Zassenhaus degree at least 2, and at most U_S further relators of Zassenhaus degree at least 3. Consequently, if there is $\tau \in (0, 1)$ such that

$$(6.4) \quad 1 - g\tau + (g + |S| + 3)\tau^2 + U_S\tau^3 < 0,$$

then the final constrained pro-2 group is infinite.

Proof. By Proposition 16, before the selected prime cuts the group H has a minimal presentation with $d = d(H)$ generators and at most $d + 4$ relators of degree at least 2.

There are $|S| - 1$ first cuts, one for each odd rational prime in S : the inert Frobenius cut, or the first square Frobenius cut for ramified and split primes. Say m' of them are non-Frattini in the intermediate quotient at the moment they are imposed. By Lemma 13, those m' cuts remove m' generators and add no relator, while the remaining $|S| - 1 - m'$ each add at most one relator of degree at least 2. After all first cuts the generator rank is $d_{\text{fin}} = d - m'$ and the number of relators of degree at least 2 is at most

$$(d + 4) + (|S| - 1 - m') = d_{\text{fin}} + |S| + 3.$$

For each of the U_S selected odd primes split in k , Proposition 20 represents the second square Frobenius condition by a relator of degree at least 3.

Next, the elementary quotient E_T/k of Lemma 17 survives all selected prime cuts, which bounds d_{fin} from below. Indeed, if p is inert in k , the unique prime of k above p has residue degree 2 over $\mathbb{Q}$, so its Frobenius acts on every rational quadratic character as the square of a rational Frobenius and is already trivial on the exponent-2 extension E_T/k . If p is ramified or split, only the squares of Frobenius elements are killed, and these too are trivial on an exponent-2 quotient. Hence the constrained group still surjects onto $\text{Gal}(E_T/k)$, and so $d_{\text{fin}} \geq g$.

Finally, the weighted polynomial of the resulting presentation is bounded above by

$$1 - d_{\text{fin}}\tau + (d_{\text{fin}} + |S| + 3)\tau^2 + U_S\tau^3.$$

For fixed $\tau \in (0, 1)$ this expression is decreasing in d_{fin} , since $\partial/\partial d_{\text{fin}} = -\tau + \tau^2 < 0$, and so replacing d_{fin} by its lower bound g can only increase it. Hence (6.4) makes the weighted polynomial negative, and Theorem 21 gives infinitude. $\square$

6.6. The Extraction of the Tower Fields. Armed with Theorem 22, we are now ready to extract the fields required by Theorem 10, following the argument of [14, Lemma 12]:

Proposition 23. *Assume (6.4) and assume $7 \in T$. Then there are totally real Galois extensions $F/\mathbb{Q}$ of arbitrarily large degree, containing*

$$\mathbb{Q}(\sqrt{q} : q \in T, \sqrt{2}),$$

such that for $K = F(i)$ the following three properties hold. (i) Every prime of F above every $p \in S$ splits in K/F . (ii) The extension K/F is unramified at all finite primes, and the discriminant parameters are

$$(\Delta_K/\Delta_F)^{1/[F:\mathbb{Q}]} = \sqrt{16D} \quad \text{and} \quad \sqrt{|\Delta_K|} = (\sqrt{16D})^{[F:\mathbb{Q}]}.$$

(iii) The local parameters may be taken as

$$e(p) = \begin{cases} 4, & p = 2, \\ 2, & p \in T, \\ 1, & p \notin T \cup \{2\}, \end{cases} \quad \text{and} \quad f(p) \leq 2 \quad (p \in S).$$

Proof. By Theorem 22, the constrained pro-2 group is infinite, and it surjects onto the finite elementary quotient $\text{Gal}(E_T/k)$. Consequently, the group has finite quotients of arbitrarily large order that still retain E_T/k . Let F'/k be the corresponding finite Galois extensions: they are totally real, unramified at all finite primes other than $\mathfrak{q}$, have dyadic local image inside $A_{\mathfrak{q}}$, and satisfy the selected prime residue degree conditions.

Let F'^* be the conjugate of F' under the non-trivial automorphism of $k/\mathbb{Q}$, and set

$$F = F'F'^*.$$

The prime $\mathfrak{q}$ is the unique dyadic prime of k , so the dyadic two ray condition is stable under conjugation, and the selected prime conditions are visibly stable. Thus $F/\mathbb{Q}$ is Galois, totally real, of arbitrarily large degree, contains E_T , and satisfies the same local bounds.

We compute the local parameters. At odd rational primes, F/k is unramified, so $e(p) = 2$ for $p \in T$ (ramified in k) and $e(p) = 1$ for odd $p \notin T$. At $p = 2$, the extension $k/\mathbb{Q}$ has ramification index 2, the retained $\sqrt{2}$ ray gives one further ramified quadratic step above $k_{\mathfrak{q}}$, and since the dyadic local image lies in $A_{\mathfrak{q}}$ there is no further dyadic ramification. Hence $e(2) = 4$, and the unramified part of $A_{\mathfrak{q}}$ bounds the dyadic residue degree by 2. For odd $p \in S$ the imposed Frobenius conditions bound the residue degree of p in F by 2, and thus $f(p) \leq 2$ for all $p \in S$.

We now set $K = F(i)$ and verify the splitting condition (i). If $p \equiv 1 \pmod{4}$, then p splits already in $\mathbb{Q}(i)$, and hence every prime of F above it splits in K/F . If $p \not\equiv 1 \pmod{4}$ is odd, the admissibility condition (6.1) provides $q \in T$ with p inert in $\mathbb{Q}(\sqrt{q})$. Since $\sqrt{q} \in F$, the residue degree of p in $F/\mathbb{Q}$ is even, so the local field at any prime of F above p contains the quadratic unramified extension $\mathbb{Q}_p(i)$ of $\mathbb{Q}_p$, and p splits in $F(i)/F$. At $p = 2$, the hypothesis $7 \in T$ gives $\sqrt{7} \in F$. Since $-7 \equiv 1 \pmod{8}$ is a square in $\mathbb{Q}_2$, we get $\mathbb{Q}_2(\sqrt{7}) = \mathbb{Q}_2(\sqrt{-1})$, so every dyadic completion of F contains i and the dyadic primes split in K/F .

For (ii), note $k(i) = k(\sqrt{-D})$ and the biquadratic discriminant identity

$$\text{disc}(\mathbb{Q}(\sqrt{D}, i)) = (4D) \cdot (-4) \cdot (-D) = (4D)^2,$$

by the conductor-discriminant formula, so the relative discriminant of $k(i)/k$ is trivial and $k(i)/k$ is unramified at all finite primes. Its base change K/F is then also unramified at all finite primes.

Finally we compute the root discriminant of F . The tower above the fixed field $k(\sqrt{2})$ is unramified at all finite primes, so $\text{rd}(F) = \text{rd}(k(\sqrt{2}))$, and $k(\sqrt{2}) = \mathbb{Q}(\sqrt{D}, \sqrt{2})$. For $D \equiv 3 \pmod{8}$,

$$\text{disc}(\mathbb{Q}(\sqrt{D})) = 4D, \quad \text{disc}(\mathbb{Q}(\sqrt{2})) = 8, \quad \text{and} \quad \text{disc}(\mathbb{Q}(\sqrt{2D})) = 8D,$$

and the conductor-discriminant formula gives

$$\text{disc}(\mathbb{Q}(\sqrt{D}, \sqrt{2})) = (4D) \cdot 8 \cdot (8D) = 256D^2.$$

Thus $\text{rd}(F) = (256D^2)^{1/4} = \sqrt{16D}$. Since K/F is unramified at all finite primes, we have that $(\Delta_K/\Delta_F)^{1/[F:\mathbb{Q}]} = \text{rd}(F) = \sqrt{16D}$ and $\sqrt{|\Delta_K|} = \text{rd}(F)^{[F:\mathbb{Q}]}$, which proves (ii) and completes the proof of the theorem. $\square$

7. THE FINITE CERTIFICATE

In this section we fix the finite data underlying Theorem 1, namely the set T , the selected set S , and the weights $k(p)$, and we carry out the verification of the criterion (6.4) and the evaluation of the master inequality. Throughout, T denotes the set of the

first 38 odd primes and $D = \prod_{q \in T} q$. We found the data by a computer search over candidate sets T of between 30 and 50 small odd primes, combined, for each candidate, with an exact optimization of S , the weights, and the radius R . Our first goal will be to certify the analytic input $\log B_{E_{39}}(\lambda) < 0.09$ used in the master inequality.

7.1. The E_{39} Louboutin Certificate. Put

$$E_{39} = \mathbb{Q}(\sqrt{q} : q \in T, \sqrt{2}),$$

a multiquadratic field of degree 2^{39} . Proposition 23 guarantees that every tower field F contains E_{39} , and so we may apply Theorem 10 with $E = E_{39}$. The point of using the full multiquadratic field E_{39} , rather than a smaller subfield, is that each ramified prime $q \in T$ and the prime 2 then contribute negative correction terms in the expression (7.1) for the degree normalized $\log \zeta_{E_{39}}$.

We begin with the local data of E_{39} :

Lemma 24. *The field E_{39} has degree 2^{39} , and its local factors have the following form. At $p = 2$ the completion is $\mathbb{Q}_2(\sqrt{3}, \sqrt{5}, \sqrt{2})$, with $e = 4$ and $f = 2$. At each $q \in T$ we have $e = 2$ and $f = 2$, where the residual quadratic direction is witnessed by some $q' \in T$ with $\left(\frac{q'}{q}\right) = -1$, listed for every q in the second table of the Appendix. At every other prime p the Frobenius has order at most 2, and p splits completely in E_{39} if and only if $\left(\frac{2}{p}\right) = 1$ and $\left(\frac{q}{p}\right) = 1$ for all $q \in T$. Consequently, for $s > 1$,*

$$(7.1) \quad \begin{aligned} \log \zeta_{E_{39}}(s)^{1/2^{39}} &= \frac{1}{2} \log \zeta(2s) - \frac{1}{4} \sum_{q \in T} (-\log(1 - q^{-2s})) \\ &\quad - \frac{3}{8} (-\log(1 - 2^{-2s})) + R_{39}(s), \end{aligned}$$

where

$$R_{39}(s) = \sum_{p \text{ split completely in } E_{39}} \operatorname{artanh}(p^{-s}).$$

Proof. We first treat the dyadic place. The square classes of odd rational primes in $\mathbb{Q}_2^\times / \mathbb{Q}_2^{\times 2}$ are determined by their residues modulo 8 and are generated by the classes of 3 and 5. Since $3, 5 \in T$, the dyadic completion is $\mathbb{Q}_2(\sqrt{3}, \sqrt{5}, \sqrt{2})$. The class 5 gives the unramified quadratic direction and the classes 3 and 2 give total ramification index 4, so that $e = 4$ and $f = 2$.

For an odd ramified prime $q \in T$, ramification comes only from $\sqrt{q}$, and any witness $q' \in T$ with $\left(\frac{q'}{q}\right) = -1$ forces a residual unramified quadratic direction, so $e = f = 2$. At primes outside $2T$ the multiquadratic extension is unramified with elementary abelian Galois group, so Frobenius has order 1 or 2, and complete splitting is exactly the simultaneous vanishing of all the quadratic characters, namely $\left(\frac{2}{p}\right) = \left(\frac{q}{p}\right) = 1$ for all $q \in T$.

We now compute the degree normalized Euler factors. A generic unramified prime with Frobenius of order 2 contributes local exponent $\frac{1}{ef} = \frac{1}{2}$, giving the factor $(1 - p^{-2s})^{-1/2}$ and in total the term $\frac{1}{2} \log \zeta(2s)$ up to the corrections below. A completely split prime contributes $(1 - p^{-s})^{-1}$ instead, and

$$\frac{(1 - p^{-s})^{-1}}{(1 - p^{-2s})^{-1/2}} = \exp(\operatorname{artanh}(p^{-s})),$$

which accounts for $R_{39}(s)$. A ramified odd $q \in T$ has normalized local exponent $\frac{1}{ef} = \frac{1}{4}$, a deficit of $\frac{1}{4}$ against the generic $\frac{1}{2}$, and the dyadic exponent is $\frac{1}{ef} = \frac{1}{8}$, a deficit of $\frac{3}{8}$. This yields (7.1). $\square$

Armed with Lemma 24, we are now ready to prove the required bound:

Proposition 25. *With $x_0 = 3 \cdot 10^{-11}$, $s_0 = 1 + x_0$, and $\lambda = \sqrt{16D}$, we have that*

$$\log B_{E_{39}}(\lambda) < 0.09.$$

Proof. We evaluate the infimum (4.1) at the single point $x = x_0$ and bound the five resulting pieces. Each numerical bound below is a strict upper bound obtained from evaluation with interval arithmetic.

First, a high precision evaluation of the explicit part of (7.1) gives

$$\frac{1}{2} \log \zeta(2s_0) - \frac{1}{4} \sum_{q \in T} (-\log(1 - q^{-2s_0})) - \frac{3}{8} (-\log(1 - 2^{-2s_0})) < 0.0887019.$$

Second, we bound the initial segment of $R_{39}(s_0)$. Complete splitting in E_{39} implies complete splitting in the subfield E_{30} generated by the square roots of the first thirty odd primes, so a deterministic enumeration of the primes up to 10^{13} that split completely in E_{30} gives

$$\sum_{\substack{p \leq 10^{13} \\ p \text{ split completely in } E_{39}}} \operatorname{artanh}(p^{-s_0}) \leq \sum_{\substack{p \leq 10^{13} \\ p \text{ split completely in } E_{30}}} \operatorname{artanh}(p^{-s_0}) < 7.34 \cdot 10^{-11}.$$

Third, we cover the range from 10^{13} to $8D$ by a Selberg upper bound sieve [7, 8] in dyadic blocks, applied within the two admissible residue classes $p \equiv \pm 1 \pmod{8}$. If $C(Y)$ denotes the number of completely split primes in $(Y, 2Y]$, then for every $Q \geq 1$ we have that

$$C(Y) \leq \frac{2(Y/8 + 1 + Q^2)}{H_{38}^*(Q)},$$

where

$$H_{38}^*(Q) = \sum_{\substack{d \mid \prod_{q \in T} q \\ d \leq Q}} \prod_{q \mid d} \frac{q+1}{q-1}.$$

With $Y_j = 10^{13} \cdot 2^j$, $Q_j = \lfloor 0.315 \lfloor \sqrt{Y_j} \rfloor \rfloor$, and $0 \leq j \leq 178$, an exact meet-in-the-middle evaluation of the sums $H_{38}^*(Q_j)$ gives

$$\sum_{j=0}^{178} \frac{2(Y_j/8 + 1 + Q_j^2)}{H_{38}^*(Q_j)} \operatorname{artanh}(Y_j^{-s_0}) < 0.00006066.$$

Fourth, for the tail $Y_j > 8D$ the Brun-Titchmarsh theorem of Montgomery and Vaughan [11] gives

$$\sum_{j \geq 179} C(Y_j) \operatorname{artanh}(Y_j^{-s_0}) < 2 \cdot 10^{-10}.$$

Fifth, the archimedean factor satisfies

$$\frac{x_0}{2} (\log \lambda - \log \pi) + \log \Gamma(1 + x_0/2) < 1.2 \cdot 10^{-9}.$$

Adding the five bounds, we obtain

$$\log B_{E_{39}}(\lambda) < 0.0887019 + 7.34 \cdot 10^{-11} + 0.00006066 + 2 \cdot 10^{-10} + 1.2 \cdot 10^{-9} < 0.09. \quad \square$$

7.2. The Certificate Data. We now turn to the discrete data of the certificate. Every discrete assertion below is verified by a finite computation in exact integer or rational arithmetic, and a verification script accompanies the source of this paper. Explicitly,

$$T = \{3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, \\ 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167\},$$

the set of the first 38 odd primes. It satisfies

$$|T| = 38, \quad D = \prod_{q \in T} q \equiv 3 \pmod{8}, \quad \text{and} \quad \#\{q \in T : q \equiv 3 \pmod{4}\} = 21,$$

so the hypotheses of section 6 hold (including $7 \in T$), and $g = |T| = 38$.

For an odd prime $p \notin T$, we call p *split* or *inert* according as $\left(\frac{D}{p}\right) = 1$ or -1 , and we call an odd prime *admissible* if it satisfies (6.1). We define S to consist of the prime 2, every odd prime $p \leq 283$, every admissible split prime $p \in (283, 1973]$, and every admissible inert prime $p \in (283, 2267]$. For this T , every odd prime in these two intervals is admissible. The largest selected prime is 2267.

The weights are

p	2	3	5	7	11	13, 17, 19	$23 \leq p \leq 53$	$59 \leq p \leq 283$	otherwise
$k(p)$	19	12	8	6	5	4	3	2	1,

where the ranges refer to primes of S in those ranges. The complete table, including splitting status and admissibility witnesses, is given in the Appendix.

We summarize the discrete half of the certificate:

Proposition 26. *Let T be the set of the first 38 odd primes, and let S and the weights $k(p)$ be as defined above. Then*

$$|S| = 314 \quad \text{and} \quad U_S = 114,$$

and the weighted Golod-Shafarevich polynomial of Theorem 22,

$$P(\tau) = 1 - 38\tau + (38 + 314 + 3)\tau^2 + 114\tau^3 = 1 - 38\tau + 355\tau^2 + 114\tau^3,$$

satisfies

$$P\left(\frac{6}{115}\right) = -\frac{101}{1520875} < 0.$$

In particular the criterion (6.4) holds for (T, S, k) .

Proof. Each assertion is a finite computation in exact integer or rational arithmetic from the definitions above. The type counts are

type	dyadic	ramified	inert	split
count	1	38	161	114

and the weight histogram is

k	1	2	3	4	5	6	8	12	19
$\#\{p \in S : k(p) = k\}$	253	45	8	3	1	1	1	1	1.

Every odd selected prime is either $1 \pmod{4}$ or carries the inertness witness $q \in T$ displayed in the Appendix. The minimum of P on $(0, 1)$ occurs at

$$\tau = 0.05220818647952869 \dots \quad \text{with} \quad P(\tau) = -6.68471161139 \cdot 10^{-5} \dots,$$

and $\frac{6}{115}$ is a nearby exact rational witness, which gives the displayed value. $\square$

Theorem 22 and Proposition 23 thus supply the required family of fields for Theorem 10.

7.3. The Master Inequality Evaluation. To prove Theorem 1, what remains is to evaluate the master inequality.

Proof of Theorem 1. The criterion (6.4) holds by Proposition 26, and $7 \in T$, and so Proposition 23 supplies totally real Galois extensions $F/\mathbb{Q}$ of arbitrarily large degree containing E_{39} such that the CM fields $K = F(i)$ are Galois, every prime of F above every $p \in S$ splits in K/F , the discriminant parameters are $(\Delta_K/\Delta_F)^{1/d} = \lambda = \sqrt{16D}$ and $\sqrt{|\Delta_K|} = \lambda^d$, and the local parameters $e(p)$ and $f(p) \leq 2$ are as displayed there. That is, the hypotheses of Theorem 10 hold with $E = E_{39}$ and $\lambda = \sqrt{16D}$.

For the selected prime weights, using $f(p) = 2$ for all $p \in S$,

$$(7.2) \quad \Sigma := \sum_{p \in S} \frac{\log(k(p) + 1)}{4e(p)} = 56.2819843775269 \dots,$$

and

$$(7.3) \quad \log A = \sum_{p \in S} \frac{k(p) \log p}{2e(p)} = 1093.4272954355824 \dots$$

The dyadic root discriminant parameter is

$$\lambda = \sqrt{16D}, \quad \text{so that} \quad \log(\lambda^2) = 154.01230121116885 \dots$$

Any $R > 1/2$ gives a valid bound in Theorem 10, and we choose the numerically optimal

$$R = 2.688292831485591 \dots$$

Inserting the rigorous bound $\log B_{E_{39}}(\lambda) < 0.09$ of Proposition 25 into (5.1) and (5.2) yields

$$N(R) > 37.8867825632423 \quad \text{and} \quad M(R) < 1057.332065023008.$$

Here

$$\log \left(\frac{\lambda}{2A} \right) = -1017.114292 \dots,$$

so the thickening term $\lambda/(2A)$ is negligible at the displayed precision, although we evaluate the exact expression in (5.2). Consequently

$$\frac{N(R)}{M(R)} > 0.0358324.$$

Taking $\varepsilon = N(R)/M(R) - 0.0358324 > 0$ in Theorem 10 gives arbitrarily large finite sets $U \subset \mathbb{R}^2$ with at least $(\#U)^{1.0358324}$ ordered unit distance pairs, which proves the theorem. $\square$

APPENDIX: THE SELECTED PRIME CERTIFICATE

The following table lists the complete selected prime set S of section 7. For each $p \in S$ it records the weight $k(p)$, the ramification index $e(p)$ of p in the tower fields F , and the status of p : dyadic ($p = 2$), ramified ($p \in T$), or split or inert in $Q_0 = \mathbb{Q}(\sqrt{D})$ according to the Kronecker symbol $\left(\frac{D}{p}\right)$. The witness column certifies admissibility (6.1): it is blank when $p \equiv 1 \pmod{4}$, and otherwise it lists a prime $q \in T$ with $\left(\frac{q}{p}\right) = -1$, while in the dyadic row the entry 5 records the unramified dyadic square class of (6.2).

Table 1: The selected prime certificate for T the first 38 odd primes.

p	$k(p)$	$e(p)$	status	witness
2	19	4	dyadic	5
3	12	2	ramified	5
5	8	2	ramified	
7	6	2	ramified	3
11	5	2	ramified	7
13	4	2	ramified	
17	4	2	ramified	
19	4	2	ramified	3
23	3	2	ramified	5
29	3	2	ramified	
31	3	2	ramified	3
37	3	2	ramified	
41	3	2	ramified	
43	3	2	ramified	3
47	3	2	ramified	5
53	3	2	ramified	
59	2	2	ramified	11
61	2	2	ramified	
67	2	2	ramified	3
71	2	2	ramified	7
73	2	2	ramified	
79	2	2	ramified	3
83	2	2	ramified	5
89	2	2	ramified	
97	2	2	ramified	
101	2	2	ramified	
103	2	2	ramified	3
107	2	2	ramified	5
109	2	2	ramified	
113	2	2	ramified	
127	2	2	ramified	3
131	2	2	ramified	17
137	2	2	ramified	
139	2	2	ramified	3
149	2	2	ramified	

Continued on next page

p	$k(p)$	$e(p)$	status	witness
151	2	2	ramified	3
157	2	2	ramified	
163	2	2	ramified	3
167	2	2	ramified	5
173	2	1	inert	
179	2	1	inert	7
181	2	1	inert	
191	2	1	inert	7
193	2	1	inert	
197	2	1	split	
199	2	1	inert	3
211	2	1	split	3
223	2	1	split	3
227	2	1	split	5
229	2	1	split	
233	2	1	split	
239	2	1	split	7
241	2	1	inert	
251	2	1	split	11
257	2	1	split	
263	2	1	inert	5
269	2	1	split	
271	2	1	split	3
277	2	1	inert	
281	2	1	split	
283	2	1	inert	3
293	1	1	inert	
307	1	1	split	3
311	1	1	inert	11
313	1	1	inert	
317	1	1	inert	
331	1	1	inert	3
337	1	1	inert	
347	1	1	inert	5
349	1	1	split	
353	1	1	split	
359	1	1	split	7
367	1	1	split	3
373	1	1	inert	
379	1	1	inert	3
383	1	1	inert	5
389	1	1	split	
397	1	1	inert	
401	1	1	inert	
409	1	1	inert	
419	1	1	inert	11
421	1	1	inert	
431	1	1	inert	7
433	1	1	split	

Continued on next page

p	$k(p)$	$e(p)$	status	witness
439	1	1	split	3
443	1	1	inert	5
449	1	1	inert	
457	1	1	split	
461	1	1	inert	
463	1	1	inert	3
467	1	1	inert	5
479	1	1	inert	13
487	1	1	inert	3
491	1	1	split	7
499	1	1	inert	3
503	1	1	split	5
509	1	1	inert	
521	1	1	split	
523	1	1	inert	3
541	1	1	inert	
547	1	1	inert	3
557	1	1	split	
563	1	1	inert	5
569	1	1	inert	
571	1	1	split	3
577	1	1	inert	
587	1	1	inert	5
593	1	1	inert	
599	1	1	inert	7
601	1	1	inert	
607	1	1	split	3
613	1	1	inert	
617	1	1	split	
619	1	1	inert	3
631	1	1	split	3
641	1	1	inert	
643	1	1	split	3
647	1	1	split	5
653	1	1	split	
659	1	1	inert	7
661	1	1	split	
673	1	1	split	
677	1	1	split	
683	1	1	split	5
691	1	1	split	3
701	1	1	split	
709	1	1	inert	
719	1	1	split	11
727	1	1	split	3
733	1	1	split	
739	1	1	inert	3
743	1	1	split	5
751	1	1	split	3

Continued on next page

p	$k(p)$	$e(p)$	status	witness
757	1	1	inert	
761	1	1	inert	
769	1	1	inert	
773	1	1	split	
787	1	1	split	3
797	1	1	inert	
809	1	1	inert	
811	1	1	split	3
821	1	1	inert	
823	1	1	inert	3
827	1	1	inert	5
829	1	1	inert	
839	1	1	inert	11
853	1	1	split	
857	1	1	inert	
859	1	1	split	3
863	1	1	split	5
877	1	1	inert	
881	1	1	inert	
883	1	1	inert	3
887	1	1	inert	5
907	1	1	split	3
911	1	1	split	7
919	1	1	inert	3
929	1	1	split	
937	1	1	split	
941	1	1	inert	
947	1	1	inert	5
953	1	1	split	
967	1	1	split	3
971	1	1	inert	11
977	1	1	inert	
983	1	1	split	5
991	1	1	inert	3
997	1	1	inert	
1009	1	1	split	
1013	1	1	split	
1019	1	1	inert	7
1021	1	1	inert	
1031	1	1	split	7
1033	1	1	inert	
1039	1	1	split	3
1049	1	1	inert	
1051	1	1	inert	3
1061	1	1	split	
1063	1	1	inert	3
1069	1	1	inert	
1087	1	1	split	3
1091	1	1	inert	17

Continued on next page

p	$k(p)$	$e(p)$	status	witness
1093	1	1	inert	
1097	1	1	inert	
1103	1	1	inert	5
1109	1	1	inert	
1117	1	1	split	
1123	1	1	inert	3
1129	1	1	split	
1151	1	1	inert	13
1153	1	1	inert	
1163	1	1	split	5
1171	1	1	inert	3
1181	1	1	split	
1187	1	1	split	5
1193	1	1	split	
1201	1	1	inert	
1213	1	1	inert	
1217	1	1	inert	
1223	1	1	inert	5
1229	1	1	inert	
1231	1	1	inert	3
1237	1	1	split	
1249	1	1	inert	
1259	1	1	inert	11
1277	1	1	inert	
1279	1	1	split	3
1283	1	1	inert	5
1289	1	1	split	
1291	1	1	split	3
1297	1	1	split	
1301	1	1	split	
1303	1	1	split	3
1307	1	1	inert	5
1319	1	1	split	13
1321	1	1	inert	
1327	1	1	split	3
1361	1	1	split	
1367	1	1	inert	5
1373	1	1	inert	
1381	1	1	split	
1399	1	1	split	3
1409	1	1	split	
1423	1	1	split	3
1427	1	1	split	5
1429	1	1	inert	
1433	1	1	inert	
1439	1	1	inert	7
1447	1	1	split	3
1451	1	1	split	7
1453	1	1	inert	

Continued on next page

p	$k(p)$	$e(p)$	status	witness
1459	1	1	split	3
1471	1	1	inert	3
1481	1	1	inert	
1483	1	1	inert	3
1487	1	1	split	5
1489	1	1	inert	
1493	1	1	inert	
1499	1	1	split	7
1511	1	1	split	11
1523	1	1	inert	5
1531	1	1	split	3
1543	1	1	split	3
1549	1	1	split	
1553	1	1	inert	
1559	1	1	inert	17
1567	1	1	inert	3
1571	1	1	inert	11
1579	1	1	split	3
1583	1	1	split	5
1597	1	1	split	
1601	1	1	split	
1607	1	1	inert	5
1609	1	1	split	
1613	1	1	inert	
1619	1	1	split	7
1621	1	1	inert	
1627	1	1	split	3
1637	1	1	inert	
1657	1	1	inert	
1663	1	1	inert	3
1667	1	1	inert	5
1669	1	1	split	
1693	1	1	split	
1697	1	1	inert	
1699	1	1	inert	3
1709	1	1	inert	
1721	1	1	inert	
1723	1	1	split	3
1733	1	1	inert	
1741	1	1	inert	
1747	1	1	inert	3
1753	1	1	split	
1759	1	1	inert	3
1777	1	1	inert	
1783	1	1	inert	3
1787	1	1	inert	5
1789	1	1	split	
1801	1	1	split	
1811	1	1	split	19

Continued on next page

p	$k(p)$	$e(p)$	status	witness
1823	1	1	split	5
1831	1	1	inert	3
1847	1	1	inert	5
1861	1	1	inert	
1867	1	1	inert	3
1871	1	1	inert	7
1873	1	1	inert	
1877	1	1	inert	
1879	1	1	split	3
1889	1	1	split	
1901	1	1	split	
1907	1	1	split	5
1913	1	1	split	
1931	1	1	split	13
1933	1	1	inert	
1949	1	1	inert	
1951	1	1	inert	3
1973	1	1	split	
1979	1	1	inert	17
2003	1	1	inert	5
2029	1	1	inert	
2039	1	1	inert	7
2053	1	1	inert	
2063	1	1	inert	5
2081	1	1	inert	
2083	1	1	inert	3
2087	1	1	inert	5
2129	1	1	inert	
2137	1	1	inert	
2141	1	1	inert	
2143	1	1	inert	3
2203	1	1	inert	3
2221	1	1	inert	
2267	1	1	inert	5

The next table certifies the local data at the ramified primes. For each $q \in T$ it lists a witness $q' \in T$ with $\left(\frac{q'}{q}\right) = -1$, certifying the residual quadratic direction at q used in Lemma 24, so that $e = f = 2$ at every $q \in T$ in E_{39} .

q	q'	q	q'	q	q'
3	5	5	3	7	3
11	7	13	5	17	3
19	3	23	5	29	3
31	3	37	5	41	3
43	3	47	5	53	3
59	11	61	7	67	3
71	7	73	5	79	3
83	5	89	3	97	5
101	3	103	3	107	5
109	11	113	3	127	3
131	17	137	3	139	3
149	3	151	3	157	5
163	3	167	5		

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